



— SACHIN GOYAL —
THAPAR INSTITUTE OF ENGINEERING
AND TECHNOLOGY, PATIALA

Financial Risk Intelligence Platform

AI/NLP risk signals from news and social media, driving tactical index rebalancing and regulator-grade portfolio stress testing.

Methodology

Problem & Approach

The Problem

Market-moving information exists in unstructured formats. Human interpretation is:

- Slow: lagging behind real-time market movements
- Subjective: vulnerable to individual bias
- Limited: restricted by human operating hours

Our Approach

- Specialist models: FinancialBERT for calibrated sentiment, a small LLM for per-entity understanding
- Backtesting: benchmarked against 15 years of real news data and market reactions

Core Outputs

- Sentiment: scaled -1 to 1 (96.7% accuracy)
- Event class: categorized across 6 primary labels
- Impact score: calibrated 1-10 based on 19,872 historical events



System Design Architecture

One engine, three layers, and two downstream decision-making modules



- **Ingestion Layer**

Aggregates data from GDELT (15m live news), 91k FNSPID headlines, and 10.5k tweets.

- **AI/NLP Risk Engine**

Executes a pipeline involving entity matching, FinancialBERT sentiment analysis, Qwen3-4B LLM processing, event classification, and impact calculation.

- **Events & Decisions**

Performs 24h event clustering to feed into volatility-adjusted index tilting and scenario stress/CET1 flow modules.

- **Operational Serving**

FastAPI backbone supporting 14 endpoints and a React dashboard with 47 verified tests.

Implementation Highlights

Choices we can defend, and the evidence behind each.

- **FinancialBERT for bulk sentiment**

Achieved 96.7% accuracy on held-out data with 30ms latency on CPU.

- **LLM for per-entity reading**

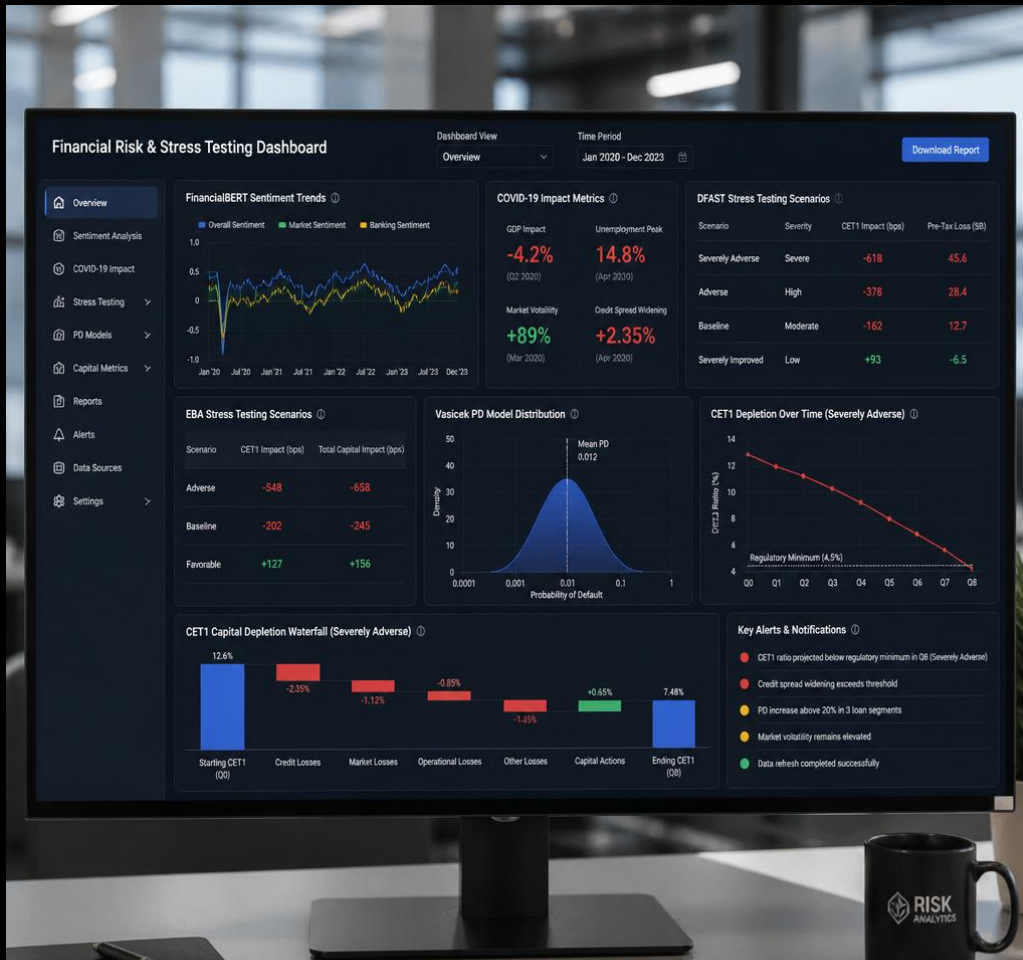
Outperformed the shared-score baseline: 72.4% vs 59.8% on SEntFiN human labels.

- **Impact calibrated on real data**

Validated against 19,872 events, showing firm-specific news drives higher alpha.

- **Regulator-grade stress testing**

Compliant with DFAST/EBA standards, featuring 2,081 bps COVID depletion modeling.



Our methodology ensures robust, data-backed performance compliant with industry stress-testing standards.

KEY RESULTS: 15 Years of Empirical Evidence

News Sentiment vs. Market Reality

Sentiment Accuracy

96.7%

FinancialBERT; bake-off of 6 models

Directional Hit Rate

56.2%

vs. 53-55% benchmark

Credit-Event Reaction

+24.9

bps CAR; $t = 2.96$, significant at 1%

Impact Score (IC)

+0.098

calibrated severity + VIX regime

Portfolio Alpha

+0.9

pts vs. cap-weight, net of costs

Risk Management

-10.4%

vol-targeted variant at 20.7% vol

Data Integrity

Derived from 78k+ articles and 10.5k tweets; no synthetic data used.

Calibrated Precision

Credit parameters stress-tested against 1.35M real Lending Club loan records.

Metrics reflect actual market history, not theoretical marketing projections.

Domain Impact

Transforming Workflow Efficiency and Regulatory Compliance



- **Workflow Efficiency**

Compresses industry-trusted risk patterns from weeks of manual effort into seconds of automated processing.

- **Intelligence & Signaling**

Automates news-to-signal extraction by combining industry-standard intelligence with real-time novelty and corroboration.

- **Regulatory Alignment**

Mirrors supervisory playbooks like CCAR/DFAST and EBA/ICAAP using transition matrices and Vasicek PD modeling.

- **Strategic Capital Output**

Delivers actionable CET1 depletion figures in basis points, purpose-built for wholesale banking and ratings analysis.

Limitations & Next Steps

Assessing performance gaps and future development path

Model Limitations



Current system issues include illustrative shock calibration, rule-based logic limitations, a modest 0.065 impact IC, and trailing performance against equal-weight benchmarks in 2022.

Roadmap Advancements



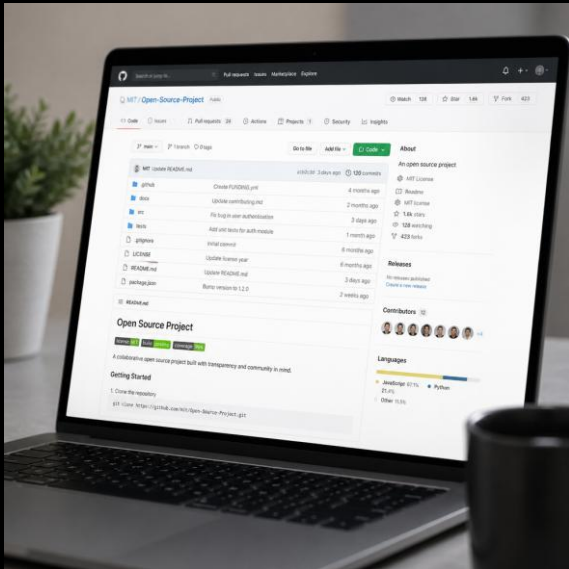
Planned upgrades include intraday event windows, fine-tuned event classifiers, embedding-based deduplication, and a transition to Kafka-based streaming architecture.

Market Expansion



Scaling coverage from the current 14 names to broader Indian markets, specifically targeting Crisil indices.

Resources & Transparency



All code, evidence artifacts, and measurement scripts are accessible via our public repository under an MIT license.